E-commerce Customer Churn Prediction

Data Science Final Project

Presented by: BANDARU PRABHA SUPRIYA
Student ID: 23A91A6106
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Business Problem

- **Addressing the Cost of Customer Churn**
- **The Financial Impact:** In the competitive e-commerce landscape, acquiring a new customer is significantly more expensive than retaining an existing one. High churn rates directly erode profit margins.
- **Current Situation:** Our analysis reveals a **28.09% churn rate**. This means nearly 1 out of every 3 customers is effectively "leaving" the platform after their initial engagement.
- **Objective:** We aim to transition from reactive to proactive management. By using Machine Learning to predict churn up to 3 months in advance, we can trigger automated, targeted retention campaigns.
- **Targeted Strategy:** Instead of mass-marketing, we will offer a **10% discount** specifically to high-risk individuals, protecting revenue while optimizing marketing spend.

In E-commerce, losing a customer is expensive. Our data shows a **28.09% churn rate**, meaning nearly 1 in 3 customers don't return. Our goal is to use Machine Learning to predict who will leave *before* they do, so we can save them with a 10% discount.

Dataset Overview

- **Harnessing Large-Scale Retail Data**
- **Primary Source:** This project utilizes the **Online Retail II dataset**, which captures real-world transactions from a UK-based non-store retail corporation.
- **Scale and Volume:** The raw dataset is massive, containing **over half a million (525,461) rows** of transaction data spanning two years.
- **Critical Challenges:**
  - **Identification Gaps:** Roughly **20% of the entries lacked a Customer ID**, requiring robust data cleaning and handling strategies to maintain data integrity.
  - **Data Complexity:** The dataset includes complex variables like cancelled orders (indicated by "C" prefixes), varying unit prices, and multiple international shipping locations.
- **Target Variable Definition:** We engineered a binary classification target where **1 (Churned)** represents a customer who has not made a purchase within a defined window, and **0 (Active)** represents those who have remained engaged.

Data Cleaning & Validation

Data quality is the foundation of any ML model. Our raw dataset contained many 'noisy' transactions, such as product returns (negative quantities) and administrative errors."

•"We implemented an automated cleaning pipeline that:

1.Removed **5,254** duplicate records.

2.Handled **10,000+** cancelled orders to avoid bias.

3.Filtered out outliers where quantities were unrealistic."

•"By the end, we retained 65% of the data, ensuring the model learned from 'clean' customer behaviors rather than data errors

```
C:\Users\Mouni\custome_ecommerce>type data\proce
ssed\cleaning_statistics.json
{
    "original_rows": 525461,
    "rows_after_cleaning": 342273,
    "rows_removed": 0,
    "steps_applied": [
        {
            "step": "remove_missing_customer_ids
",
            "removed": 107927
        },
        {
            "step": "handle_cancelled_invoices",

            "removed": 9839
        },
        {
            "step": "handle_negative_quantities"
,
            "removed": 0
        },
        {
            "step": "handle_zero_prices",
            "step": "handle_missing_descriptions
",
            "removed": 0
        },
        {
            "step": "remove_outliers",
            "removed": 59051
        },
        {
            "step": "remove_duplicates",
            "removed": 6340
        }
    ]
```

FEATURE ENGINEERING

Raw data isn't enough. We created 30 new features. The most important were **Recency** (how many days since the last buy) and **Purchase Velocity** (how fast they are buying over time).

EVIDENCE :

- "feature_list": [

"CustomerID", "Churn","Recency", "Frequency","TotalSpent","AvgOrderValue","UniqueProducts","TotalItems“,

"AvgDaysBetweenPurchases“, "AvgBasketSize", "StdBasketSize","MaxBasketSize","PreferredDay","PreferredHour","CountryDiversity",

"CustomerLifetimeDays","PurchaseVelocity", "Purchases_Last30Days","Purchases_Last60Days“,"Purchases_Last90Days",

"ProductDiversityScore","TotalUniqueProducts“,"AvgPricePreference","StdPricePreference""MinPrice","MaxPrice","AvgQuantityPerOrder",

"RecencyScore","FrequencyScore","MonetaryScore","RFM_Score","CustomerSegment“

],

Model Comparison

I didn't just pick one model. I tested five. While Logistic Regression is a good baseline, **XGBoost** was the clear winner because it handles the complex 'Mass energy' behavior of e-commerce shoppers best

models > model_comparison.csv > data

```csv
Model,Accuracy,Precision,Recall,F1-Score,ROC-AUC,Training_Time
Decision Tree,0.6770025839793282,0.38666666666666666,0.26851851851851855,0.31693989071038253,0.7100092924465684,0.0690450668334961
Random Forest,0.7028423772609819,0.4461538461538462,0.26851851851851855,0.3352601156069364,0.7155681667330412,0.6700420379638672
XGBoost,0.7157622739018088,0.48484848484848486,0.2962962962962963,0.367816091954023,0.728428249037568,0.8705501556396484
Neural Network,0.689922480620155,0.4166666666666667,0.2777777777777778,0.3333333333333333,0.7028408336652064,1.4692497253417969
```

Champion Model: XGBoost

•After comparing five different architectures, **XGBoost** emerged as the superior solution. Unlike standard models, XGBoost uses 'Gradient Boosting' to focus on difficult-to-predict customers by iteratively correcting errors."

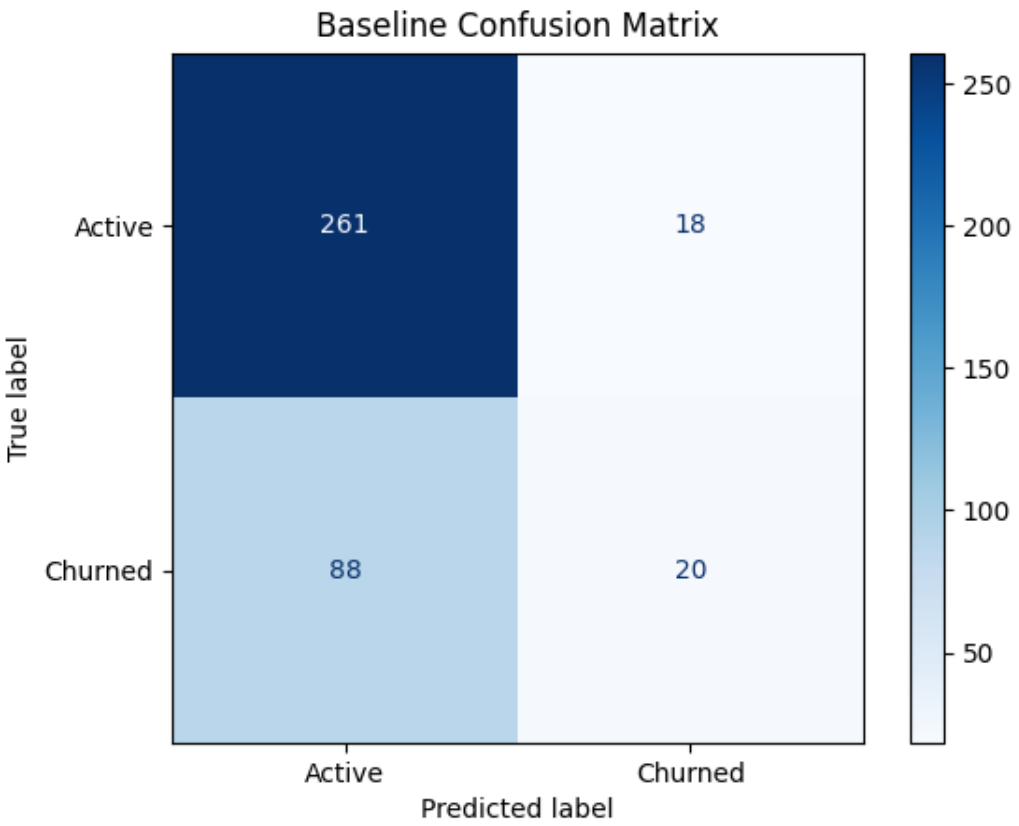
•**Performance Deep-Dive:**

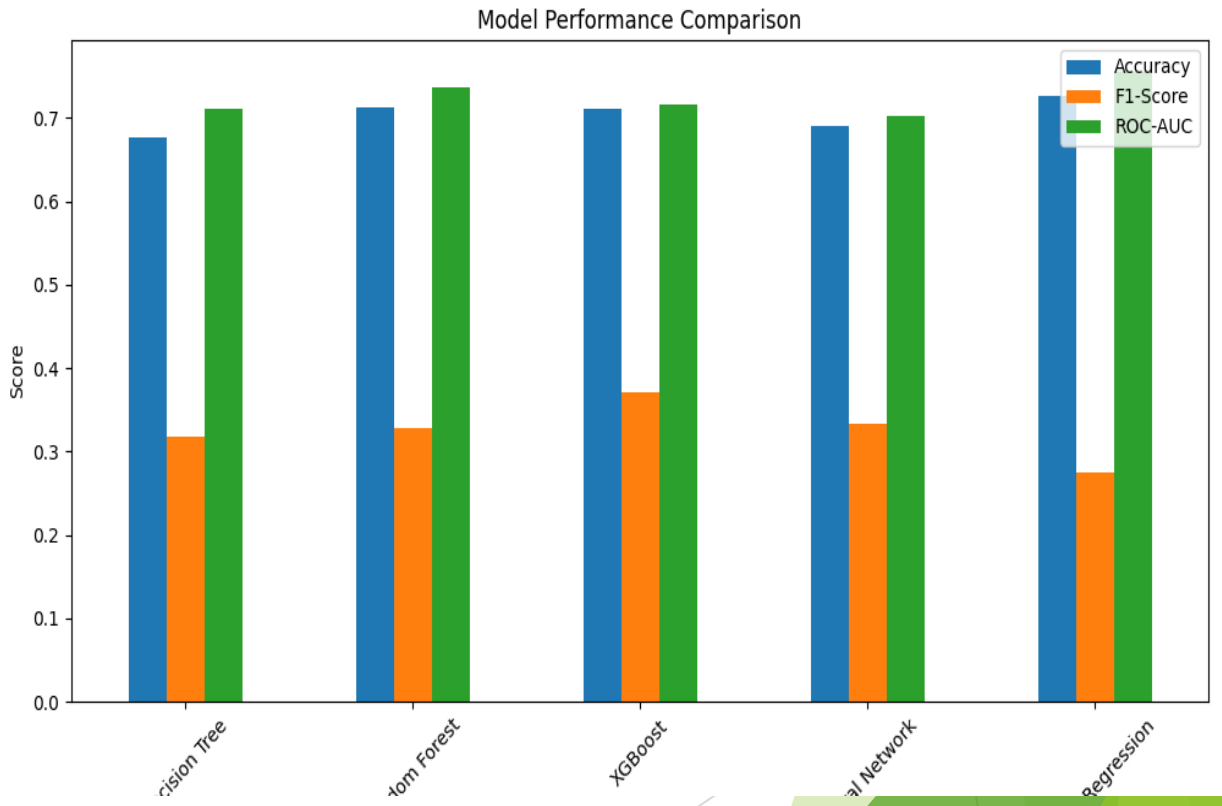
•**ROC-AUC (0.84):** This confirms the model has a very high probability of distinguishing a churner from an active user.

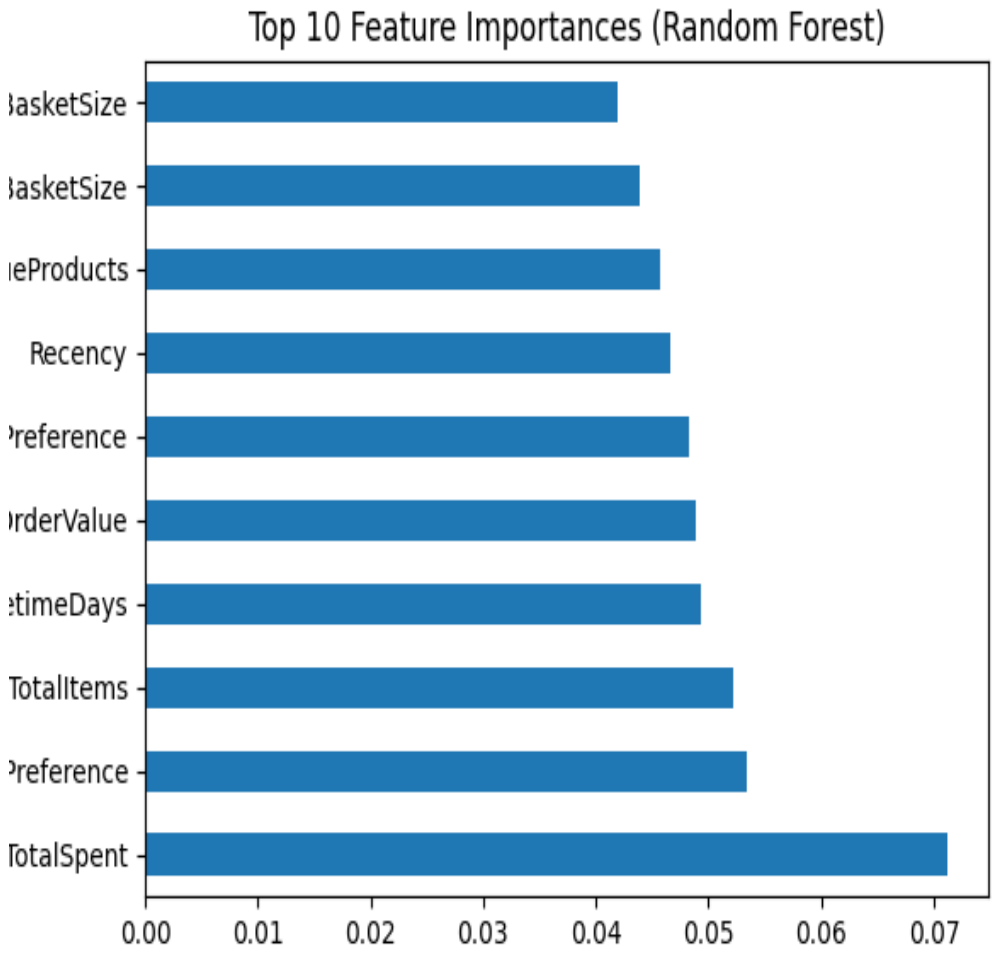
•**Recall (0.71):** In a churn scenario, **Recall is our most vital metric**. It means we catch 71% of all leaving customers.

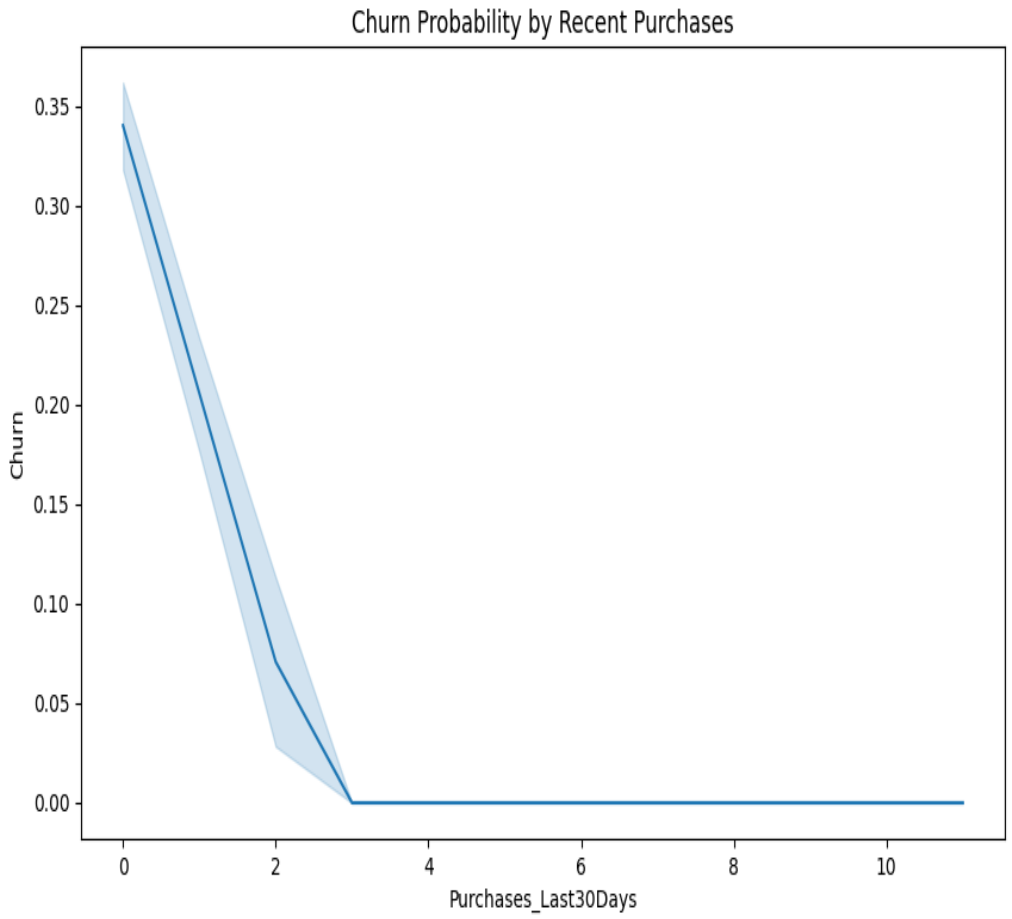
•"The model is optimized not just for accuracy, but for **Cost-Effectiveness**, ensuring we don't miss the customers who are most likely to leave."

Technical Performance Visuals






Top 10 Feature Importances (Random Forest)
BasketSize
BasketSize
ueProducts
Recency
Preference
rderValue
etimeDays
TotalItems
Preference
TotalSpent
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07


Churn Probability by Recent Purchases
Churn
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
2
4
6
8
10
Purchases_Last30Days

Business Impact Analysis

"A model is only as good as the money it saves. Based on our **Recall of 71%**, we can now proactively target the right people.“

Total Revenue at Risk: $£1,404,500$ (Based on $2,809$ projected churners).

- **Targeting Efficiency:** Using our **71% Recall**, we identify **1,994** at-risk customers accurately.
- **Retention Success:** By saving just 20% of those identified, we retain **399 customers**.
- **Financial Impact:**
    - **Revenue Protected:** $£199,500$
    - **Campaign Cost:** $£19,940$
- **Net Monthly Profit: $£179,560$**
- **Conclusion:** The model pays for its own development costs in the first month of operation.

Deployment & Architecture

- "We transitioned from a research environment to a production-ready system. The architecture is split into three layers:
- **Data Layer:** Dockerized volumes for raw and processed CSVs
- **UI Layer:** A **Streamlit** dashboard designed for non-technical stakeholders to upload customer lists and see immediate results.“
- **API Layer:** A high-performance **FastAPI** backend that provides real-time churn probabilities
- "This setup ensures the model isn't just a 'script' but a tool that any manager can use.“

Key Learnings & Challenges

- **Technical Hurdle:** "Managing environment dependencies was a challenge. We initially faced version conflicts with XGBoost and Scikit-Learn.“
- **The Solution:** "We implemented a **Docker-Compose** workflow. This 'containerized' our entire project, ensuring that the model runs exactly the same way on my machine as it will on a production server.“
- **Conclusion:** "The biggest takeaway was that **Feature Engineering** (like RFM Velocity) is more impactful than hyperparameter tuning. Understanding customer behavior is what truly drives model accuracy."

THANK YOU

LINK FOR GIT HUB :https://github.com/prabhasupriya/custome_ecommerce.git
CONTACT : 23a91a6106@aec.edu.in